

Final Project in Numerical Programming

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Course: Numerical Programming

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Transition to New Year Greeting

1. Introduction

The objective of this project is to simulate a smooth and collision-free transition of a drone swarm from an already formed handwritten configuration to a new static textual greeting, namely:

“Happy New Year!”

Unlike Previous problem, where drones start from an artificial initial grid, here the swarm begins from a nontrivial, previously computed configuration. This introduces additional constraints on stability, assignment optimality, and trajectory smoothness.

2. Mathematical Formulation

Sub-problem 2 is best treated as an IVP because:

- The transition must be smooth and collision-free
- Intermediate trajectories matter more than final-time accuracy
- Repulsion forces are active throughout the motion
- The final configuration does not need to be reached at a prescribed T

Using an IVP ensures:

- Continuous collision avoidance
- Stable convergence
- Easy chaining between sub-problems
- Robustness to different drone counts and text shapes

A BVP solver would need to:

- Handle thousands of coupled nonlinear equations
- Enforce terminal constraints that have no physical necessity
- Be sensitive to initial guesses

2.1 Inputs, Initial conditions and parameters

User Defined Parameters

Parameter	Meaning	Value
N	Number of drones	args.n_drones (150, 225, 300)
Greeting Text	Target shape definition	Happy New Year!
d	Spatial dimension	2 or 3
M	Drone mass	1.0
V_{max}	Max velocity	50
K_p	Attraction gain	3.5
K_D	Damping gain	2.0
K_{REP}	Repulsion gain	18.0
R_{SAFE}	Safety radius	1.2
Δt	Time step	0.02
Solver	Time integrator	RK4

Note: Parameter values were refined with the assistance of AI tools to improve numerical stability and visual quality. However, all final values were validated experimentally by me.

State Variables

As in Sub-problem 1, each drone i is described by:

$$x_i(t) \in \mathbb{R}^d, \quad v_i(t) \in \mathbb{R}^d$$

with $d = 2$ or 3

The global system state remains:

$$y(t) = [x_1, \dots, x_N, v_1, \dots, v_N]$$

The state at time $t = 0$ is loaded from the checkpoint produced at the end of Sub-problem 1. Therefore, previous problem determines initial conditions.

2.2 Governing Equations (IVP)

The same **initial value problem (IVP)** formulation is reused:

$$\begin{aligned}\dot{x}_i &= v_i \cdot \min\left(1, \frac{v_{max}}{\|v_i\|}\right) \\ \dot{v}_i &= \frac{1}{m} \left[k_p(T_i - x_i) + \sum_{j \neq i} f_{rep}(x_i, x_j) - k_d v_i \right]\end{aligned}$$

Where:

- T_i is the new target position for drone i in the greeting text
- The gains k_p, k_d, k_{rep} are unchanged to preserve dynamic consistency

This reuse of the same model ensures numerical continuity across sub-problems.

2.3 Collision Avoidance Model

Collision avoidance is handled using an inverse-cube repulsive force:

$$f_{rep}(x_i, x_j) = \begin{cases} k_{rep} \frac{\|x_i - x_j\|}{\|x_i - x_j\|^3}, & \text{if } \|x_i - x_j\| < R_{safe} \\ 0, & \text{otherwise} \end{cases}$$

Key properties:

- Strong short-range repulsion
- Zero force beyond safety radius
- Smooth behavior for numerical stability

This guarantees collision-free trajectories even for large N .

3. Greeting Text Generation

3.1 Motivation

The greeting text must:

- Be clearly readable
- Be uniformly populated by drones
- Allow a smooth morphing from the handwritten shape

To satisfy these requirements, the greeting text is procedurally generated rather than loaded from an image, as

- Procedural text guarantees topological consistency
- No noise, no threshold tuning
- Deterministic results across runs

3.2 Text Rendering and Skeletonization

`generate_text_points` method ensures that text is rendered using OpenCV with a handwriting-style font:

`cv2.FONT_HERSHEY_COMPLEX`

Steps:

1. Render text into a binary mask
2. Apply morphological skeletonization
3. Extract contours from skeleton
4. Sample points uniformly by arc length

This produces a clean centerline representation of the greeting text, ensuring:

- Stroke fidelity
- Uniform drone spacing
- Visual continuity with Sub-problem 1

3.3 World Coordinate Mapping

Sampled image points are transformed to world coordinates using the same mapping as before:

$$(x, y)_{\text{image}} \rightarrow (x, y)_{\text{world}}$$

Using the same transformation guarantees:

- Identical scaling
- No visual distortion between sub-problems

4. Drone–Target Assignment

At the start of Sub-problem 2, the drone swarm is already arranged in the handwritten signature formation. To prevent trajectory crossings and unnecessary motion during the transition to the “*Happy New Year!*” greeting, the drone–target assignment problem is solved again.

The assignment is formulated as:

$$\min_{\pi} \sum_{i=1}^N ||x_i - T_{\pi(i)}||^2$$

where x_i is the current position of drone i , T_j represents a target point on the greeting text, and π is a permutation defining a one-to-one mapping between drones and targets.

The problem is solved using the Hungarian algorithm, which guarantees a globally optimal assignment minimizing total squared distance. Each drone is thus assigned a unique target point. The attractive force driving the motion is then defined as:

$$F_{att,i} = K_P (x_i^{\text{target}} - x_i)$$

Implementation Details:

- Target positions are from
`generate_text_points(args.greeting_text, args.n_drones, args.dim)`
- Assignment from
`assign_targets_hungarian(curr_pos, target_pts)`
- The proportional gain K_P controls convergence speed and smoothness

Computational Complexity

The Hungarian algorithm has a worst-case time complexity of $\mathcal{O}(N^3)$, where N is the number of drones. Since the assignment is computed only once at the start of Sub-problem 2, its computational cost remains acceptable for swarm sizes up to several hundred drones and does not dominate the overall simulation time.

Key Benefits

- Globally optimal drone-target pairing
- Minimal total displacement during transition
- Smooth and visually coherent morphing between formations

This assignment step is essential. Naive index-based assignment leads to intersecting trajectories, unnecessary motion, and an increased risk of collisions, especially for complex target shapes such as text.

5. Numerical Integration

5.1 Time Integration

The ODE system is solved using **4th-order Runge–Kutta (RK4)**:

$$y_{n+1} = y_n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

Chosen because:

- Local truncation error $\mathcal{O}(\Delta t^4)$
- Stable for smooth force fields
- Handles stiff attraction–repulsion balance well

The time step:

$$\Delta t = 0.02$$

gives small enough error to be stable for the chosen gains.

5.2 Stability Considerations

It is important to note that the classical RK4 method is not A-stable. In particular, for the linear test equation RK4 is stable only within a bounded region of the complex plane.

Therefore, in this project stability is ensured by:

- Velocity saturation $v_{\max}$
- Damping term $-k_d v$
- Repulsion force clipping
- Small integration step

As a result:

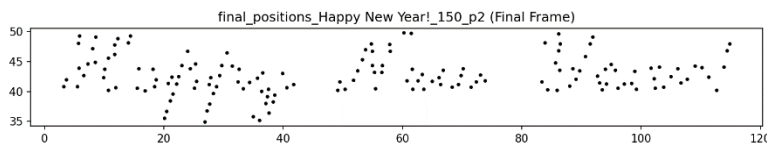
- Velocities decay smoothly to zero
- The final greeting shape is static and stable
- No drone collisions occur

6. Test Cases and Observations

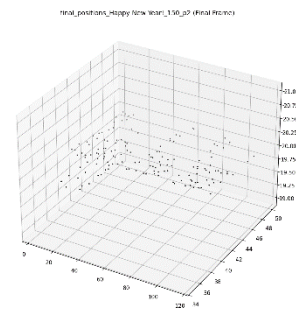
The video results demonstrate that all drones were successfully transitioned from the handwritten signature to the “Happy New Year!” text. However, achieving an optimal visual outcome required adjusting the number of drones, which represents the main limitation of this approach.

All test cases were evaluated in both two- and three-dimensional settings. In both cases the system performed reliably, indicating that a two-dimensional formulation is fully sufficient for this sub-problem and does not compromise the quality of the result.

Test Case 1: N = 150 drones (Insufficient density)



*you can see the video in desired outputs file (2d or 3d)
with name: drone_show_Happy New Year!_150_p2.mp4*

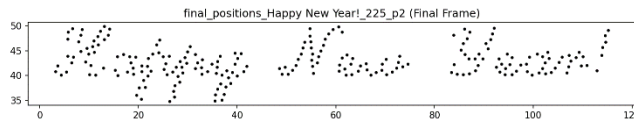


Observation:

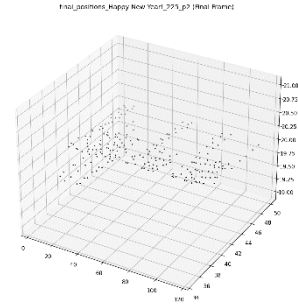
- Letters appear broken, with visible gaps, especially in curved strokes.
- Thin parts of characters are under-represented.
- Drones converge quickly but settle into a sparse point cloud.
- The Hungarian assignment still works optimally, but optimal assignment cannot compensate for missing degrees of freedom.

This test demonstrates that the limiting factor here is **sampling density**, not numerical instability or control design.

Test Case 2: N = 225 drones (Optimal balance)



*you can see the video in desired outputs file (2d or 3d)
with name: drone_show_Happy New Year!_225_p2.mp4*



Observation:

- The greeting text is **clearly readable and well-shaped**.
- Letter strokes are continuous with uniform spacing.
- No visible clustering or collisions occur.
- Convergence is smooth and visually pleasing.

Explanation:

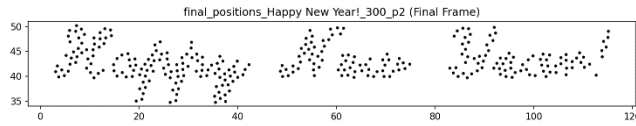
At 225 drones, the arc-length sampling density closely matches the geometric complexity of the greeting:

- Each character stroke is represented by enough points to appear continuous.
- The repulsive force maintains separation without distorting the shape.
- The proportional attraction term $K_p(T_i - x_i)$ dominates without causing overshoot.

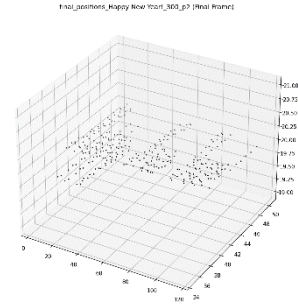
This configuration achieves an effective balance between:

- spatial resolution,
- collision avoidance,
- numerical stability,
- computational cost.

Test Case 3: N = 300 drones (Over-sampling)



*you can see the video in desired outputs file (2d or 3d)
with name: drone_show_Happy New Year!_300_p2.mp4*



Observation:

- The greeting is very smooth and dense.
- Some drones compete for nearly identical target points.
- Computation time increases noticeably.

Explanation:

With 300 drones, the system becomes over-constrained relative to the available path length:

- Many drones are assigned to target points that are very close to each other.
- Repulsive forces activate more frequently due to reduced inter-drone spacing.
- The increased number of pairwise interactions $O(N^2)$ amplifies small numerical fluctuations.

7. Conclusion

Sub-problem 2 successfully demonstrates a smooth and collision-free transition of a drone swarm from a handwritten signature to a static greeting text using an IVP-based formulation. By reusing the same dynamical model, collision avoidance strategy, and numerical integrator as in Sub-problem 1, numerical continuity and stability across stages were preserved.

The use of arc-length-based sampling for text generation, combined with a globally optimal Hungarian assignment, ensured smooth trajectories and visually coherent morphing between formations. Test cases with varying swarm sizes showed that the quality of the final greeting is primarily limited by sampling density rather than numerical stability or control design. An intermediate drone count ($N = 225$) provided the best balance between shape fidelity, stability, and computational cost.

The experiments further confirmed that a two-dimensional formulation is sufficient for this task, as extending the motion to three dimensions did not provide a noticeable qualitative advantage. Overall, Sub-problem 2 validates the robustness, scalability, and modularity of the proposed swarm control framework.

8. AI Usage Declaration

Artificial Intelligence tools (chatgpt.com) were used during the development of this project in a supportive and transparent manner. Specifically, AI assistance was employed for:

- Improving the clarity and structure of the written report
- Polishing code style and adding explanatory comments
- Giving me ideas of which input images would success and which not
- Refining parameter choices for numerical stability and visualization quality.
- Identifying and fixing implementation bugs
- Assisting in the formulation and refinement of numerical methods and modeling choices
- Suggesting parameter ranges to improve numerical stability and visual quality

All algorithmic decisions, numerical models, and final implementations were designed, validated, and executed by me. The AI tools were used as an assistant (in no case as an agent) and did not replace independent problem solving or implementation.